CHAPTER SIX

# TOWARD REAL UTOPIAS

> There is a sense in which all technical activity contains an inherent tendency toward forgetfulness.
>
> —*Langdon Winner,* Autonomous Technology[1]

> What good is having the right to sit at a lunch counter if you can't afford to buy a hamburger?
>
> —*Martin Luther King, Jr.*[2]

> Let's say the research goes on. The drugs are developed. They're very expensive, and what good has it done you if you can't access them?
>
> —*Senator Sheila Kuehl*[3]

**ON DEC 1, 2004**, the television host and political commentator Rachel Maddow hosted a radio program, *Unfiltered*, in which she dedicated the regular segment called "Burying the Lead" to a California ballot measure that she suspected very few people knew about. In this case, Maddow was not referring to California's Proposition 71 but Proposition *72*, a bill that would require large- and medium-size companies to provide health insurance to their employees and pay at least 80 percent of the cost. At the time of Maddow's reporting, the ballots were still being counted, but the voting seemed to be turning in favor of the Yes on 72 campaign, which ultimately came up short, with 49.2 percent yes votes.

Perhaps guilty of burying the lead as well, I have waited until nearly the end of this book to fully question what a "right to stem cell research" means in a context in which basic healthcare access is not yet a guaranteed right in the United States. For whose bodies have we codified a

2013. Stanford University Press.
All rights reserved. May not be reproduced in any form without permission from the publisher, except fair uses permitted under U.S. or applicable copyright law.

EBSCO Publishing: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES 713483; Ruha Benjamin; People's Science : Bodies and Rights on the Stem Cell Frontier Account:ehost.

right to develop novel cures? Presumably, the de facto constituency of Prop. 71 are those people who already have access to existing treatments, because for those who do not, a right to basic healthcare would necessarily come before a futuristic investment like stem cell research. This tension between elite and populist priorities led one observer to frame the larger implications of Prop. 72 as "the most important sleeper on the ballot this year":

> Without knowing much about it, people don't realize they may be voting on the hope of national health care reform. Prop. 72 [may] sound boring, but [it's] really one of the most important battles in the country this November. It's David vs. Goliath on health care reform. . . . Why should you care? Because huge companies see it as a litmus test for national health care reform, and they want to stop it. . . . Can you imagine a world where people weren't chained to jobs because of insurance? Can you imagine actually turning back the tide of healthcare inflation? Can you imagine standing up to giant corporation machines like Wal-Mart and finally winning? That's what's at stake, not just here but for the country, on Prop 72.[4]

As for those "giant corporations," top donors to the No on 72 campaign included the California Restaurant Association, Wal-Mart, and McDonald's, as well as a number of other national chain stores and restaurants,[5] while top donors to the Yes on 72 campaign included the Service Employees International Union, the California Teachers Union, the Food and Commercial Workers Union, and the California Healthcare Association.[6] This showdown reflects what pundits worry is the rising "class warfare" in the United States, as if the policies that supported the accumulation of wealth in the hands of a smaller and smaller subset of the population over the last several decades were not in themselves an assault upon the opportunities and life chances of the majority of Americans.

Exemplifying the age-old struggle between the strong and the weak, a biotech firm actually named Goliath worked to mobilize opposition against Prop. 72, threatening that "in the end, the very people that

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

[Prop. 72] was attempting to help—employees without health care—will not only *not* have health insurance, they won't have jobs."[7] Another No on 72 advocate offered the following scenario, repeated in various forms in the days leading up to the November 2 election:

> Consider a minimum-wage employee with minimal health insurance coverage. Since his salary cannot be cut any further by law, terminating him would be the only way for a business to cope with the costs of this mandate. As a direct result of Proposition 72, this employee would not only fail to see any improvement in his health coverage, he would lose his job and the minimal coverage he had previously enjoyed. Likewise, a business with 25 employees would be faced with a choice of shouldering the unsustainable burden of Proposition 72 and facing bankruptcy or laying off 6 of its employees to become exempt. It doesn't take a CEO to realize the route this hypothetical business will take.[8]

Other biotech companies followed suit, joining in a campaign against the "economic catastrophe" of Prop. 72 while throwing all of their support behind Prop. 71.[9] One after another, they expressed concern about the country's healthcare crisis but insisted that Prop. 72 was *not* the answer. Bill Goodrich, president and CEO of the United Agribusiness League (UAL), the voice of the agricultural industry, argued that "[a]n answer to the uninsured population clearly needs to be found. But with California's current economic difficulties, now is not the time to burden our businesses with employer-mandated healthcare coverage."[10] The public was warned of the pitfalls of "government-run healthcare," despite the fact that numerous labor, medical, and consumer groups tried to counter that false charge.[11] The California Black Women's Health Project, among others, clarified that "Proposition 72 is not a government-run healthcare system. It is private insurance, paid for by employers for employees. Prop. 72 provides a state purchasing pool for employers who choose not to purchase health insurance directly. Over one million uninsured Californians will stop depending on taxpayer-paid care for the uninsured and start getting private insurance paid for by employers."[12]

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

While I am not convinced that Prop. 72 would have had a sustainable, long-term impact for those who most need access to quality healthcare or that it could avoid the threat of increasing unemployment due to employers' cutting costs, its defeat alongside the passage of Prop. 71 does reflect an underlying tension between elite and populist "rights": investment in future medical breakthroughs versus expanding access to present public goods. In light of other public spending priorities, one observer explained it this way:

> While $300 million per year [the budget of the stem cell agency] is only one third of one percent (0.35 percent) of the State's $86 billion General Fund budget, it is nonetheless a misplaced commitment when State Medi-Cal and Workmen's Compensation funds are being drastically cut. For example, the State Legislative Analyst has recommended that funding for visits to physicians and treatment centers could save $196.5 million per year by limiting visits to 10 per year. . . . But the self-perpetuating California Center for Regenerative Medicine is standing in the way. We can no longer afford luxury jobs programs for biomedical professionals for hypothetical research which is already amply funded by both the private sector and the National Institutes of Health, while medically needy people are in need of resources for care in California.[13]

But while many Prop. 72 opponents expressed concern for "medically needy people" and some even granted that the state has a responsibility to ensure that they receive healthcare, ideas for how to achieve this were scarce. Instead, proposals to build in accountability for equitable healthcare access, specifically in the context of the stem cell initiative, were repeatedly shot down by most members of the Independent Citizens' Oversight Committee and the agency's most loyal supporters.

For example, in the year following the passage of Prop. 71, California senator Deborah Ortiz was at the forefront of sponsoring legislation to regulate the stem cell agency.[14] Stem cell advocates routinely criticized these legislative efforts as power-hungry meddling, as when physicist and Nobel Prize winner Paul Berg commented that "this is a whole new

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

structure laid on top of the state government, which has the responsibility for giving away three billion dollars. And as you might imagine, the legislators are very, very envious and jealous."[15]

The most controversial bill was SB 401, which sought to tighten regulation around CIRM's intellectual property policy. CIRM opposed the bill, in part because "language directed towards increasing the share of revenues to be allocated to the state from Prop. 71's future successes may have the effect of discouraging applications for research grants," and because "participation by important research groups and entities could be [negatively] impacted by tying what they can charge for their costs, should a product (drugs, therapies, etc) be developed, to federal Medicaid pricing levels."[16] Reacting to the proposed changes, Americans for Cures circulated a letter to supporters that called Deborah Ortiz "an ongoing threat to Proposition 71," explaining that

> [i]n a stream of legislative acts, Senate Bill 18, Senate Constitutional Amendment 13 and most recently Senate Bill 401, the Senator has attempted to impose crippling restrictions on the research she claims to champion. Rather than allowing the CIRM time to develop policy and standards, Senator Ortiz attempted to impose her own legislative controls on the new agency barely one month after its approval by the voters. Her policies, if implemented, would be disastrous for the new stem cell research program. SB 401 will stunt progress in stem cell research. . . . Above all, SB 401 ignores the single most important element in stem cell research: the patients and their families.[17]

Senator Ortiz also seemed to have patients and their families in mind—specifically those who would find themselves unable to afford stem cell treatments if expedience outweighed equity in the implementation of the initiative. At a June 6, 2005, meeting of the Independent Citizens' Oversight Committee,[18] Ortiz explained that

> [w]e have a huge challenge every year to serve the poor in our healthcare programs, in our constantly shrinking budgets, and that

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

> has been sort of the promise we presented to the voters in Prop. 71. It's that policy I think we remain in disagreement on how best to effectuate. . . . And then we balance that objective and that value against the need to assure that essential medical research is not unreasonably hindered by the intellectual property agreements. . . . What I think is the highest and most important objective in my efforts is to assure a stream of treatments to our poor in California.[19]

For many board members and stem cell advocates, the senator's demands were seen as premature, if not also overly restrictive and burdensome. Robert Klein requested the legislature to back off and give the stem cell agency the "ability to innovate and try and figure out how we can make these programs affordable."[20] Don Reed seconded this, suggesting that "greatness needs time to grow."[21] A few weeks earlier, without Ortiz present, Reed had been even more insistent:

> We can be polite as we want; but if we're not clear and blunt and outspoken, she'll win, because the people in there, first of all, they like power. They like oversight. They want control. It's in their nature to control. And if she, the expert, whom they [other members of the legislature] know was supportive [of stem cell research], if she says control is needed, their first instinct is going to be to go with her. . . . The attempt to make this affordable has been tried before. The NIH has tried it, and there's a good study, which shows that it was an utter disaster. What people came to realize is that before we can have affordable computers, we must have computers. The greater good is the benefit of this, not the small individual tinkering, which will slow the whole thing down.[22]

Another member of the public, Denise Reynolds, whose good friend was paralyzed in a biking accident, implored, "Let us not become greedy in demanding greater revenue and outcost treatments to the low-income populations."[23] ICOC member and Burnham Institute director John Reed added that "[t]o me (SB 401) is death by 1,000 cuts."[24] Similarly, David Baltimore, president of the California Institute of Technology

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

and a board member and investor in several well-known biotech companies, addressed Ortiz directly at the meeting:

> You are trying to burden this bill with a huge social problem in America, which is the disparity between the healthcare available to the poor and the healthcare available to the rich. And I am very sensitive to that and totally supportive of trying to do something about it. But if you burden an initiative whose focus is research with solving that problem, then you get yourself involved in all of these very detailed issues, and at the same time it is a snare and a delusion to believe that the resources that are going to come from this research are going to be able to pay for the needs of the poor. . . . We're talking about something that's simply not going to exist. You're worried about the return to Californians. Research is not about financial return. Research is about setting the basis for therapy. . . . I would wish that somebody with your obvious sympathy with the need for research would cheer on this group rather than tying us up so that we are unable to carry out the function that was provided for us through Proposition 71.[25]

Perhaps the idea that "research is not about financial return" was a truism forty years ago, but in the wake of the Bayh-Dole Act and the privatization of research, Baltimore's earnest appeal rings hollow. The question is not whether the research should seek to be profitable but rather *profitable for whom?* The agency was doing all it could to ensure that private biotech companies would profit from stem cell research, because they were seen as a crucial gatekeeper in the race for cures, even as the act itself recognized "the conflicting interests involved in IP [intellectual property] by providing that 'The ICOC shall establish standards that require that all grants and loan awards be subject to intellectual property agreements that balance the opportunity of the State of California to benefit from the patents, royalties, and licenses that result from basic research, therapy development, and clinical trials with the need to assure that essential medical research is not unreasonably hindered by the intellectual property agreements.'"[26]

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

When SB 401 was eventually rejected and Senator Deborah Ortiz had reached her term limit, Senator Sheila Kuehl replaced Ortiz as a primary antagonist of the agency, going on record as saying that "[b]asically, the attitude has been that they just want the Legislature to go away so they can run their own show."[27] She warned that the agency's oversight proposal was "a weak and vague standard that [was] unlikely to result in any meaningful access for the uninsured to new stem cell drugs and therapies. The proposed regulations actually take a step backwards from previous iterations, by allowing the [biotech company–sponsored] access plans" to police themselves.[28] To address this, Kuehl introduced SB 1565, a proposal that would require the agency's intellectual property standards to

> include a requirement that each grantee and the licensees of the grantee submit to the CIRM for approval a plan that will afford uninsured Californians access to any drug that is, in whole or in part, the result of research funded by the CIRM, and would require that any plan subject to that approval shall require that the grantees and licensees thereof sell drugs at a price that does not exceed any benchmark price in the California Discount Prescription Drug Program.[29]

Although it passed by a vote of 16–0, Governor Schwarzenegger eventually vetoed the bill.

Reacting to ongoing antagonism between the stem cell agency and the California legislature, consumer watchdog advocate John Simpson complained that "[p]art of the problem with this whole thing has been an ongoing attitude of, 'You're with us or you're against us and you're against stem-cell research.' That's just nonsense. It ought to be the case that you can have constructive criticism of things that a state agency does without people shouting back at you that you're against stem-cell research."[30] "When they get responsible criticism from outsiders, they kind of circle the wagons and say, 'We know science, we know best.' I think they would be much better off if they would sit down and engage."[31] In many ways, the circle-the-wagons solidarity that had helped

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

mobilize a disparate group of patient advocates, scientists, Hollywood celebrities, and biotech investors in the passage of Prop. 71 had become a liability. The zealous pro-cures and proscience campaign rhetoric that had made for effective sound-bite politics was now alienating potential allies at a time when more tempered and inclusive deliberation was required to implement the initiative.

We know that unlike his predecessor, Barack Obama has been a major supporter of stem cell research, using his first act as president to loosen federal regulations on the struggling field. Despite the red-blue polarization for which SCR is known, it is Obama's national healthcare bill that has garnered the fiercest criticism and backlash from political opponents. How then do the bioconstitutional struggles over equity and inclusion in stem cell research fit into the broader U.S. healthcare debate?

It is a challenge to address this, in part because the current U.S. health system is undergoing shifts that in many ways mirror Prop. 72's effort to mandate health insurance via employers. The Obama administration has created an imperative for individuals and families to purchase insurance, with a significant subsidy for those who make between 100 and 400 percent of the federal poverty level, alongside other major provisions. Reflecting on the impact of the current administration's health reform, *New York Times* columnist David Leonhardt reported that

> For all the political and economic uncertainties about health reform, at least one thing seems clear: The bill that President Obama signed on Tuesday is the federal government's biggest attack on economic inequality since inequality began rising more than three decades ago. . . . Much about health reform remains unknown. . . . Maybe the bill's attempts to hold down the recent growth of medical costs will prove a big success, or maybe the results will be modest and inadequate. But the ways in which the bill attacks the inequality of the Reagan era—whether you love them or hate them—will probably be around for a long time.[32]

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

Leonhardt goes on to frame the larger tension animating proponents and opponents of Obama's health reform as arriving at fundamentally different answers to "the central question that both the Reagan and Obama administrations have tried to answer—what is the proper balance between the market and the government?" While the 2010 health reform certainly shifts the balance toward the latter, its impact is as yet undetermined.

But even as we await the effects of federal health reform, the relationship between healthcare access, social inequality, and medical innovation is more apparent and can guide our thinking about public spending. An article on stem cell research and social justice pointed out that (prior to the impending reforms)

> as many as 46.6 million people in the United States lack health insurance. . . . Those without it are less likely to receive preventative care and thus to be in advanced stages of a disease once they are examined by a physician. They also tend to be sicker upon being admitted to a hospital and thus more likely to die after being admitted. In addition, even though those without health insurance tend to be poorer, they often have greater out of pocket expenses and have higher rates of bankruptcy due primarily to their medical expenses.[33]

Ethnoracial minorities, in turn, "are much more likely to be without health insurance than white Americans. In 2005, 11.3 percent of non-Hispanic whites lacked health insurance, while 19.6 percent of African Americans and 32.7 percent of Hispanics did not have it. The impact of not having health insurance on people's health and financial viability cannot be overemphasized."[34]

An analysis of National Health Statistics data found that between 1991 and 2000, almost 900,000 African American deaths could have been prevented if blacks had received the same healthcare as whites.[35] The study estimates that "technological improvements in medicine—including better drugs, devices, and procedures—averted only 176,633 deaths during the same period."[36] The authors thus questioned "the pru-

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

dence of investing billions in the development of new drugs and technologies while investing only a fraction of that amount in the correction of disparities," given that the deaths averted through improvements in technology constituted only a fraction of those that could had been saved through access to basic healthcare. The lead author of the study, Steven H. Woolf, explained that "five times as many lives can be saved by correcting disparities than in developing new treatments."[37] A second author, Otis Brawley, explained that "it's important [to note] that this is not an argument against science. . . . This is an argument that there are therapies out there that are not new that people just don't get."[38] A similar study using 2002 data found that "an estimated 83,570 excess [African American] deaths each year could be prevented in the United States if this black-white mortality gap could be eliminated."[39]

## From Center to Margin

Political philosophers and progressive educators alike have long emphasized the importance of storytelling in opening up the civic sphere to a broader range of publics. Rational decision-making, whether about health, safety, education, or fiscal responsibility, requires that the stories of otherwise marginalized groups enter and influence public discourse. Genuine empathy is a necessary feature of this discourse; impartiality is not about discarding feelings and vetting opposing claims through "dry logic." Rather, a robust civic sphere requires us to "take the position of the concrete other in order to judge problems *more* impartially."[40] The universal citizen—that abstract figure who exercises rights and bears responsibilities in the modern polity—is a myth. By this I do not mean that it is a baseless delusion, but that it is a story that we tell ourselves: an ideal that we desire but a fiction nonetheless that hides the many ways in which our current social order does not simply produce "outsiders within" society, but relies upon them.[41]

While important work in the last several decades has sought to bring marginalized perspectives into the center of public discourse and intellectual thought, the inverse process of considering major developments in the life sciences *from the periphery* is perhaps an equally vital shift. The

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

public sponsorship of stem cell research in California is part of a longer and larger story about citizenship, about how priorities are established, about how economic value and collective values intertwine, and about how the future is imagined. What we have observed in the previous chapters is how, much too often, the few speak for the many: the elite speak for the masses, projecting the desire for future treatment onto people who cannot obtain *existing* therapies which could prolong and save their lives *right now*.

Citizenship, after all, is not simply a political status so much as it is a *process*.[42] The challenge for us is to enact health and science policy in a way that acknowledges what we share and have in common while at the same time taking account of important, often life-and-death social cleavages. Rather than opening up space within which to deal critically with the differential impact of health policy and priorities on different populations, public engagement with science is typically manufactured to recognize only a certain set of prescribed differences (such as those recognized in the ten disease advocacy representatives on the ICOC) that do not challenge the underlying prioritization of innovation over equity.

## Beyond Bioconstitutional Struggles?

Finally, it is not enough to simply broaden the parameters of social struggle, allowing for a more "fair" contest between the haves and have-nots over biomedical goods (for example, the right to "fair access"), without at once querying the underlying assumptions of the struggle. Shifting attention from consumption of science and technology as the primary means of exercising citizenship, to participation as a more robust form of citizenship, in which people actively engage in the production of knowledge, forces us to contend with challenges that are even more entrenched than administering universal healthcare. In *Re-thinking Science: Knowledge and the Public in an Age of Uncertainty* (2001), Nowotny and colleagues explain that "in the eyes of many citizens, science and technology are now equated with their results and products. They have come to be seen largely as commodities, access to which should be democratically regulated and the allotment of which

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

should also be fairly distributed. Consequently, what should be produced and how it should be produced must be embraced within democratic decision making."[43]

We have inherited "a social structure that has accepted separation as the norm—the dividing of people into groups of 'we' and 'they' who fight, who compete, who negotiate, who cooperate, or who help each other from across the boundaries that define their separateness."[44] I borrow this last insight from a prescient compilation of papers called *The Lab, the Temple, and the Market*, in which an interdisciplinary group of analysts reflect upon the role of science, religion, and economic development in human emancipation and social subordination. They question the role of scholars in reifying social differences and group struggle, and ask whether

> human beings [are] doomed to be outsiders to every group except a single one, a subculture narrowly defined by nationality, ethnicity, social class, religion, and occupation. . . . This tendency reinforces, and is reinforced by, an intellectuality that sees as the hallmark of intelligence the ability to identify differences, to divide, and to relativize, all in the name of being scientific. Such an approach is a gross misrepresentation of science, for although it is true that science analyzes, it also integrates and points to underlying patterns of oneness.[45]

In particular, the work of Farzam Arbab, physicist, educator, and founder, in 1974, of the Colombia-based Fundación para la Aplicación y Enseñanza de las Ciencias (Foundation for the Application and Teaching of the Sciences, or FUNDAEC),[46] informs my thinking about how we can imagine and enact more egalitarian forms of participation in science in a way that is liberating and just. While the Colombian and Californian contexts are dramatically different on a number of fronts, they are similarly ordered by the "current arrangements that assign the ownership of modern science to small sectors of society." If alternatives to this social order are not created, we can expect a "widening of the gap between the poor and the rich," which is as well a deeply racialized cleavage.[47]

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

In California, for example, where the poverty rate is already typically higher than the rest of the nation, Latinos (22.8%) and African Americans (22.1%) have a substantially higher rate of poverty than whites (9.5%) and Asians (11.8%).[48] The life-and-death implications of racialized economic inequality in turn are nowhere more evident than in mortality rates. In California, a black man "can expect to live 68.6 years on average, which is far below the life expectancy of the average California white male, who can expect to live 75.5 years,"[49] leading a Harvard University professor of public health, Nancy Krieger, to assert in no uncertain terms that "social inequality kills."[50] Her work, among that of others, demonstrates that popular mandates and movements for social justice (for example, the Civil Rights Movement and the War on Poverty, including Medicare and Medicaid, which began closing the racialized gap between rich and poor) were the primary drivers for alleviating health inequities between 1966 and 1980.

> Starting thereafter, in 1980, with the Reagan revolution we've had 25 years of conservative policies that have sought to roll back the "Welfare State," [and] rein in big government except, of course, for military spending. And we have seen this play out in a familiar litany of deregulating industry, slashing [taxes] on the wealthy, repeatedly freezing the minimum wage, and undermining funding for public health infrastructure. . . . The net result has been widening health inequities. *So are these inequities inevitable?* The clear answer is "no." Our data show that the inequities shrank and then they have widened. *What are the alternatives?* We as a nation have to say that we have the political and social priorities, because we *do have* the economic resources to address these inequities.[51]

In sum, a large part of moving forward entails remembering—Sankofa-like—what has worked in the past rather than allowing ourselves to be hypnotized by the latest shiny innovation promising to revolutionize health. But rather than bracing ourselves for a roller coaster of progress and retrenchment corresponding to changes in political administrations, we need to tackle a much more fundamental question: not

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

simply, what kinds of policies and initiatives do we wish to undertake, but also, who are the agents of change?

Here I draw upon a new wave of sociological thinking about "real utopias," which challenges us to imagine and create institutions and social relations that are fundamentally different from the status quo.[52] As 2012 president of the American Sociological Association Erik Olin Wright describes it, a real utopia embraces the

> tension between dreams and practice. It is founded on the belief that what is pragmatically possible is not fixed independently of our imaginations, but is itself shaped by our visions. Self-fulfilling prophecies are powerful forces in history. . . . Nurturing clear-sighted understandings of what it would take to create social institutions free of oppression is part of creating a political will for radical social changes to reduce oppression. A vital belief in a utopian ideal may be necessary to motivate people to leave on the journey from the status quo in the first place, even though the likely actual destination may fall short of the utopian ideal.

He emphasizes that *real* utopias are "ideals that are grounded in the real potentials of humanity, utopian destinations that have accessible waystations, utopian designs of institutions that can inform our practical tasks of muddling through in a world of imperfect conditions for social change."[53]

In a similar vein, at the Science and Democracy lecture series at Harvard in 2010, writer and activist Arundhati Roy exhorted participants: "The first step towards reimagining a world gone terribly wrong would be to stop making war on those who have a different imagination, an imagination outside of capitalism and, for that matter, communism, an imagination which has an altogether different understanding of what constitutes happiness and fulfillment."[54] Consider that throughout the previous pages you have been introduced to many people willing to invest money and time to hone scientific techniques that might defy (what were once accepted as) laws or limits of biology as they relate to aging and illness—as when journalist Teri Somers asks us to "[i]magine cardiac

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

cells, beating in a petri dish, being used to form human tissue that might be used to replace damaged heart muscle."[55] But somehow imaginations go limp when we are confronted with social dis-order. For many people, the idea that we can defy politics as usual and channel human ingenuity toward more cooperative and inclusive forms of social organization is utterly far-fetched. Thus I am convinced that any institutional retooling must begin by first querying this faith in biological regeneration alongside our underdeveloped belief in social transformation. *If our bodies can regenerate, why do we perceive our body politic as so utterly fixed*?

That being said, I would like to spend the last pages of this book sketching a few principles for participatory science-making that, while not suggesting a fully fledged real-utopian blueprint, seek to outline several tenets that I consider crucial to any such effort. The proviso is that it would be antithetical to any such institution-building to offer directives; rather, I offer these principles as raw material for designing "technologies of humility"[56] that must ultimately be the subject of wide-ranging consultation as we create alternatives to top-down, elite-driven science and technology policy. Like Farzam Arbab, I am concerned about "the right of the masses of humanity not only to have access to information [or therapies], but to participate fully in the generation and application of knowledge."[57] To that end, I have distilled five lines of praxis from the otherwise far-reaching essays in *The Lab, the Temple, and the Market*—practices which I think are essential pillars in the construction of real-utopian designs for participatory science.

*Not Equating Winning with Being Right*. Democracy as it is currently imagined and enacted divides people "according to interest, talent, and ideology, who then 'negotiate' decisions," wherein "the purpose of each component group is to win. The means to this end are economic advantage and the mobilization of support to overwhelm the opponent. So strong is this legacy of 'he who wins is right' that it essentially determines the way justice is administered."[58] We see these dynamics at play in the Prop. 71 and 72 campaigns, as well as in the implementation of Prop. 71, wherein those who monopolize capital were best positioned to materialize their particular vision of the good life. How often did the

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

proponents of stem cell research use the ballot win to assert that "the people of California" gave them a mandate to pursue expedience over equity? Their "win" in turn bolstered their opposition to incorporating forms of equitable distribution of stem cell goods. Arbab urges us to question whether we should accept this idea that "he who wins is right" as the "crowning achievement of the evolution of collective decision-making on the planet."[59] To which I answer, emphatically and optimistically, *no*. The fierce competition we take for granted in "democratic" life, wherein those who monopolize capital wield inordinate power and influence in the name of the many, is an outmoded form of top-down governing. Even when the winner *doesn't* take all, "losers" are still left to squabble over crumbs, as when only 2–5 percent of Big Science initiative budgets is routinely allocated to investigate ethical, legal, and social issues related to the field. When public participation is staged, it often takes the form of the shallowest form of exchange, as when state agencies routinely stage "public hearings" at which speakers are allotted no more than three minutes to express their views, while final decisions are still monopolized by a select few, who often make them behind closed doors.

*Cultivating the Art of Listening over That of Manipulation*. Whether governing conventional spheres of life, such as housing or education, or more novel spheres, such as regenerative medicine, we can no longer rely upon the art of political manipulation as the taken-for-granted backdrop of decision-making. Rather, *a posture of learning* must inform and precede our collective investigations and rational analysis of options.

> Such an attitude is basically different from that of experts or highly paid consultants who generally act on the basis of a series of certainties coming from their "knowledge" or "professional experience." Such "authorities," particularly when they refuse to question their certainties, not only tend to mislead the people in whose lives they intervene, but also lose touch with the very objects of their knowledge. Because they are unable to listen, they find that their accumulated knowledge soon becomes obsolete and of little relevance to the changing realities they address.[60]

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

An example of public policy decision-making based upon thoughtful listening rather than manipulation can be found in the Interactivity Foundation's "citizen discussions."[61] As part of my fieldwork on the California stem cell initiative in 2007, I took part in a series on "Anticipating Human Genetic Technology," which started off with the expectation that those gathered were "an informed citizenry *offering impressions*, rather than a persuaded citizenry *offering opinions*."[62] A facilitator introduced a number of alternatives for thinking about the development of genetic technologies, and the group proceeded to unpack the implications and unintended consequences of pursuing each alternative; for example, we explored the concerns around and benefits of "limiting human genetic technologies," "embracing human technologies," and "balancing social and individual control."[63] We then came up with suggestions for redesigning these alternatives, followed by broader observations and questions about each one. This process of consultation was neither wholly detached nor impassioned, but it deepened participants' awareness of the many sides of a policy issue (even if we may have initially come in with a strong point of view) rather than polarizing us in the way that electoral politics do.

The posture of learning that participants assumed as we collectively explored the specifics of various institutional designs was cultivated in part through the Interactivity Foundation's deliberation tools, which were expertly applied by the facilitator. That is to say, simply putting "diverse" people in a room with the "freedom" to express their viewpoints is not an effective or sustainable alternative to top-down policy decision-making. Mere inclusion of different points of view is not a panacea for the conflict-ridden politics to which many of us are accustomed; indeed, precisely because we are accustomed to it, we are likely to fall into habitual scripts in which vying for the upper hand seems "natural." Rather, the Interactivity Foundation asks prospective discussants to help one another "think through . . . ideas," not as advocates or competitors but by "step[ping] outside their own thinking and think[ing] about issues as people from different backgrounds might see them."[64]

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

*Developing Consultative, Not Consumptive, Capacity*. In thinking through the challenge of cultivating a wider capacity for decision-making beyond the privileged realm of legislatures, consultants, and think tanks, Farzam Arbab, in *The Lab, the Temple, and the Market*, explains that

> [i]n a world all too given to twisting words to suit economic interests, the capacity to make proper technological choices could easily become synonymous with the possession of the skills of a good consumer. This is clearly not what is intended here. The type of capacity under discussion represents a complex set of attitudes, convictions, understandings, skills, and habits, all of which characterize the behavior of individuals and organizations in their daily interaction with technology.[65]

The more that people are conceived of and treated primarily as consumers and end users, the less their creative capacity to imagine and produce alternatives can be realized.[66]

The tension between consuming and producing the fruits of science appears to also animate UC Berkeley professor David Winickoff's argument that we must seek not only equitable *benefit-sharing* but also equitable *power-sharing* structures: "While benefit sharing should be applauded insofar as it attempts to submit relations of biocapital to new claims of distributive justice, the project is likely to fail both as a normative and practical matter without greater attention to issues of procedural justice."[67] Again, the Interactivity Foundation offers one model for expanding decision-making and power-sharing capacity, in which we engaged in five stages of "citizen policy discussions":

1. Describe the area of concern by developing questions
2. Generate policy possibilities that respond to those questions
3. Explore possible consequences in order to revise the policy options
4. Organize joint panel discussions between persons with professional experience relevant to the area of concern and citizens whose familiarity with the area of concern grows out of their general life experi-

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

ences outside of work (typically called "lay" persons, their experiential knowledge is no less vital to the process)

5. Create a citizen discussion report (and, I would add, items for collective action)

This is not a process that can be rushed through in one session or during a single conference, however well intentioned—and certainly not in the kinds of public hearings that are the current hallmark of "open" and "transparent" governance at California's stem cell agency. When we consider the often slow, deliberate, incremental process by which scientific knowledge is typically generated, why is it that we do not expect or implement the same kind of attentive dynamics in *social* experiments that seek to generate new knowledge? Rather, we have a deeply asymmetrical approach, in which our investment of both time and money in reengineering biological life far exceeds our collective will to transform *social* life.

*Fostering Reciprocity Between Institutions and Individuals.* It is vital that institutions cease imposing the views of a particular faction or class, whether democratically elected or not, on the rest of society. Conservation biologist and MacArthur fellow John Terborgh has argued that "[t]he public, and especially the political class, cherry-picks its science. If a scientist finds a promising treatment for AIDS or cancer, then he is a hero; if he warns about overpopulation, climate change, or toxic contamination of the environment, then he risks either being ignored or, worse, being subjected to ridicule. Such negative incentives reduce to a handful the number of scientists who are willing to speak out."[68] Thus, it is not reciprocity *for reciprocity's sake* that obliges our institutions to be more attentive to a wide range of expertise, nor is the latter simply that expertise which speeds up production of sought-after biomedical goods. Instead, an inclusive attention may produce better science in not weeding out individuals who may add valuable insights about the unintended risks or hidden possibilities of pursuing a particular course of action:

> To the extent that institutions become channels through which the talents and energies of the members of society can be expressed in

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

> service to humanity, a sense of reciprocity will grow in which the individual supports and nurtures institutions and these, in turn, pay sincere attention to the voice of the people whose needs they serve. . . . Meeting this challenge implies a *fundamental change in the process of decision-making*, both individual and collective. Today, unbridled competition, obsession with power, and the abuse of authority vitiate the way decisions are made. The process suffers from extremes: apathy or over-enthusiasm, attachment to technique or haphazardness, devotion to minutia or the propensity to deal only with abstractions. What is vitally needed is a mode of operation into which systematic learning has been woven.[69]

By applying the principle of justice at every stage of decision-making, policymakers can "avoid the pitfalls of uniformity while still respecting the exigencies of equity."[70]

*Recreating Decision-making Bodies as Learning Organizations.* "What is at stake is the transformation of the present mode of governance, based on traditional concepts of power and authority, into one shaped by a genuine posture of learning . . . [to] foster in the inhabitants of each region the capacity to make increasingly more valid choices, both individually and collectively, regarding the development, transfer, and adoption of technology."[71] The social spaces necessary for this transformation must be widespread, embracing traditional learning centers such as universities, virtual realities such as the blogosphere, and public spaces such as town halls and community centers—but also settings we do not typically associate with collective deliberation over science and technology, such as health clinics, job training centers, places of worship, farms, and factories. In all these settings, a deeper awareness must be cultivated that science and technology are *not* neutral, "but at a more fundamental level, technology carries with it an ideology and pronounces on the way individual and social life should be organized. Technological choice bears on every other choice made about the quality and direction of life in a region. It is itself an expression of values."[72] The challenges to such bottom-up engagement cannot be overstated.

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

As some of the foremost analysts of participatory science have pointed out, "Science and scientists have not been used to the [social] context speaking back, so it is not surprising that they see contextualization as a challenge to their cognitive and social authority. . . . They fear that irrationality will break through the fragile crust of scientification. There is hard evidence that such fears are exaggerated."[73]

## Forgetting to Remember

In *Autonomous Technology: Technics-out-of-Control as a Theme in Political Thought* (1978), Rensselaer Polytechnic Institute professor of political science Langdon Winner explores how many contemporary problems are due to "the widespread idea that technology is an autonomous force inducing change in society in ways beyond the control of human beings. Some celebrated the operation of this force, and others lamented it. But both groups were victims of the subtle paralysis of thought that such a belief produces in everyone who submits to it."[74] So the task of cultivating decision-making capacity and power-sharing structures cannot be directed simply at technophiles or at technophobes but at the seemingly passive middle as well. To that end, the passivity of Victor Frankenstein as depicted in *Autonomous Technology* provides a cautionary tale, for here is a person who "refuses to ponder the implications of his discovery":

> He is a man who creates something new in the world and then pours all of his energy into *an effort to forget*. His invention is incredibly powerful and represents a quantum jump in the performance capability of a certain kind of technology. Yet he sends it out into the world with no real concern for how best to include it in the human community. . . . He never moves beyond a dream of progress, the thirst for power, or the unquestioned belief that the products of science and technology are an unqualified blessing for humankind. Although he is aware of the fact that there is something extraordinary at large in the world, it takes a disaster to convince him that the responsibility is his. Unfortunately, by the time he overcomes his passivity, the consequences of his deeds have

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

> become irreversible, and he finds himself totally helpless before an unchosen fate.[75]

From one perspective, the architects of Prop. 71 can be seen as breaking away from the passivity described above, instead seeking to direct the fate of science and technology toward the speedy production of novel biomedical goods. However, as the preceding chapters demonstrate, they did so within the narrow frame of consumptive biological citizenship and by implicitly excluding a number of vital concerns related to race, gender, class, and disability recognition and redistribution.[76]

In moving forward, it is vital that we construct participatory models for the development of science and technology by drawing upon the tools developed by practitioners experimenting with decision-making and power-sharing. Evolving over the past forty-five years, the Interactivity Foundation has been honing its approach to policy discussions on topics that run the gamut from crime and punishment to human impacts on climate change, the future of childhood and higher education, immigration, as well as those related to new biotechnologies. Likewise, Farzam Arbab's FUNDAEC approaches people as potential resources, not as problems to be managed nor as beneficiaries to be served. The Interactivity Foundation and FUNDAEC, among other organizations, avoid offering people "prepackaged" solutions on which they simply vote; rather, the process of *elaborating alternatives* is itself central to their method.

A number of analysts have drawn upon the classical idea of the "agora" as a civic platform or process to bring together different perspectives about science and technology. It is "the domain (in fact, many domains) in which contextualization occurs . . . neither state nor market, neither exclusively private nor exclusively public, the agora is the space in which societal and scientific problems are defined, and where what will be accepted as a 'solution' is being negotiated."[77] Some describe it as the emergence of a "new social contract" between science and society, wherein reliable knowledge must give way to socially robust knowledge, and scientific autonomy gives way to scientific accountability.[78] Others characterize a triple helix model of university, government, and indus-

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

try relations, in which "the distinction between laissez faire and active-state intervention [is] obsolete," replaced by recombination of these three arenas, to which I would add a fourth, civic sphere that proves more challenging to integrate on an "equal level in the network."[79] In the actual implementation of this new social contract, public engagement has taken the form of onetime "consensus conferences," at which stakeholders gather to explore the various issues related to a particular development in science and technology. While this is certainly an improvement over the more exclusive and top-down forms of decision-making, a more radical shift is necessary still, in which engagement is more pervasive, ongoing, and informal, so that the capacity for thoughtful decision-making, and ultimately power-sharing, is not relegated to a participatory elite.

An example of cultivating decision-making power can be found at UCLA's Center for Society and Genetics' initiative at the King Drew High School in Watts, Los Angeles. As part of this program, students explored topics ranging from DNA forensics, pharmacogenomics, genetic discrimination, newborn screening, near-relative DNA forensic testing, gender testing, and designer babies, among other topics. Through debates, discussion questions, games, plays, videos, and writing exercises over the course of an academic year, students became familiar with the science behind these developments. But, and this is crucial, they also acquired skills to thoughtfully engage the ways in which science and technology acquire meaning for them personally, as well as for their families and communities.[80] Crucial to such initiatives is that participants consider a range of alternatives with respect to how various biotechnologies can be integrated in social life to bring about social equity.

While this and similar programs are currently configured as "science education outreach," in part to secure grants that are framed as such, in reality this form of capacity-building has as much to do with civic empowerment. On the surface, this type of program is not a radical departure from existing programs we typically find at elite prep schools at which students are provided with opportunities to build and experiment with science and technology in ways that prepare them to be drivers of innovation later in life. What distinguishes UCLA's program,

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

in part, is that it engages young people from a predominantly African American and Latino working-class neighborhood, a demographic that comprises 99 percent of the student body.[81] Empowering young people early on, not only with the requisite scientific know-how but also with the skills to critically think about and discuss the social, economic, and political issues that are inherent in the innovation process, is perhaps one of the most important arenas for cultivating participatory science. It places the onus on educational institutions—and, and even more, on public and private funders—to recognize the importance of this broader skill set, which is less a matter of "community outreach" and "science education" than it is one of empowering young people, and the wider communities of which they are a part, to become producers of knowledge. This is not just scientific and technical knowledge but also ethical and sociopolitical knowledge that both complements and complicates conventional thinking about innovation as always progressive.

These efforts to cultivate participatory science are all the more pressing because of the way that science and technology are busily reconstituting peoples' biological as well as social lives. The bioconstitutional struggles we have discussed reveal the tension between the incredible value and commercialization of human tissue for research and the simultaneous dilemmas posed by human agency and interests. One study found that "the root of public ambivalence [about commercialization] seems to lie in (i) notions of justice and fairness about private profit being made through public exploitation, and (ii) a perceived lack of control in terms of governance."[82] For this reason, stem cell research and human genetics are spurring creative proposals for reconstituting the relationship between science and society so as to move us beyond the constitutional amendments and electoral politics that characterize Prop. 71 and a host of copycat legislation.

For example, David Winickoff proposes a legal architecture for

> partnership governance [that] would empower participants to exert a share in distributive decision-making in return for contributing to the economic and social capital of the project. . . . Partnership

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.

> governance seeks to go further than existing mechanisms of "community consultation," by implementing control rights at the level of the research participant collective. . . . [B]uilding an architecture *ex ante* for the legitimate representation of research participants, before challenges are encountered, is likely to be a good investment.[83]

Alluding to schemes like Prop. 71's royalty proposal, Winickoff points to the shortcomings of "benefit sharing [that] attempts to stitch a distributive norm at the seam of market and gift economies."[84]

By contrast, participatory science-making must repurpose existing initiatives: taking them apart, discarding features that stand in the way of civic inclusion, and redesigning them to be more in line with "partnership governance." At the same time, to transform the social fabric out of which future initiatives will be designed, we must continue to create more fundamental changes. Among these, empowering young people must be among our foremost priorities. As the most sought-after consumer market, they must likewise be at the forefront of casting off the expectation that they are simply consumers of scientific and technical goods produced by an innovative elite. Rather, they, as do we all, have both the right and responsibility to conceive new kinds of goods—material and social—that don't simply benefit us personally but contribute to our collective well-being as well.

EBSCOhost: eBook Collection (EBSCOhost) printed on 3/23/2026 5:07:19 PM UTC via STANFORD UNIVERSITY LIBRARIES. All use subject to https://www.ebsco.com/terms-of-use.